

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

JNANA SANGAMA, BELAGAVI-590015, KARNATAKA



PROJECT REPORT ON

DESIGN AND IMPLEMENTATION OF EVENT DRIVEN AND TIME TRIGGERED USING RTOS COMMUNICATION

Submitted in partial fulfilment of the requirements for the V semester

Bachelor of Engineering

in

Electronics and Communication Engineering

of

Visvesvaraya Technological University, Belagavi.

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CERTIFICATE

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We also declare that, to the best of our knowledge and belief, the work reported here does not form part of any other report on the basis of which a degree or award was conferred on an earlier occasion on this by any other student.

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Place: Bengaluru

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ABSTRACT

The proposed project focuses on the design and implementation of an RTOS-based smart hospitality and patient care monitoring system using Event-Driven and Time-Triggered communication techniques. In hospitals and healthcare environments, timely medicine delivery is essential for patient safety and recovery. However, due to heavy workloads and emergency situations, nurses may occasionally forget or delay providing medicines to patients at the scheduled time. To address this issue, the proposed system introduces an intelligent communication mechanism between patients and nurses using real-time embedded technology.

The system operates using two communication approaches. In the Time-Triggered mechanism, medicine reminders are generated automatically at predefined intervals according to the patient's prescription schedule. The RTOS scheduler continuously monitors the timing and sends notifications or alerts to the nurse when medicine delivery time arrives. This ensures periodic monitoring and improves accuracy in medicine administration.

The Event-Driven mechanism is activated whenever the patient requires immediate attention or when the scheduled medicine is not provided on time. In such situations, the patient can send a request or alert message to the nurse through a communication interface such as a button, mobile application, or wireless module. Once the event is triggered, the RTOS immediately processes the request and sends a notification to the nurse station for quick response. This approach improves responsiveness and reduces delays in patient care.

The proposed system uses RTOS features such as task scheduling, inter-task communication, message queues, semaphores, and interrupt handling to manage multiple operations efficiently in real time. The integration of event-driven and time-triggered communication enhances reliability, patient safety, and healthcare efficiency while reducing human errors in hospitals. The system can be implemented using embedded controllers, IoT modules, sensors, and wireless communication technologies for smart hospital applications.

This project provides a cost-effective and scalable solution for modern healthcare systems by ensuring timely medicine delivery, improving nurse-patient communication, and enhancing overall hospital management through real-time automation and monitoring.

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CHAPTER 1

INTRODUCTION

GOAL:

In modern hospitals, nurses are responsible for managing multiple patients and ensuring medicines are provided at the correct time. Due to busy schedules, emergency situations, and heavy workloads, medicine delivery may sometimes be delayed or forgotten, which can affect patient safety and healthcare quality. To solve this issue, an RTOS-based smart patient care system using Event-Driven and Time-Triggered communication is proposed. In the time-triggered approach, automatic reminders are generated at scheduled intervals to notify nurses about medicine timings, while in the event-driven approach, patients can send alert messages to nurses if medicines are delayed or immediate assistance is needed. The system improves real-time communication, reduces human errors, and enhances hospital management efficiency through reliable task scheduling and real-time monitoring.

PROBLEM STATEMENT:

- Nurses handle multiple patients simultaneously, increasing workload and stress.
- Medicines may be delayed or forgotten during busy hospital operations.
- Manual monitoring systems are less efficient and prone to human error.
- Patients may not have an immediate way to remind nurses about missed medicines.
- Delayed medicine administration can affect patient recovery and safety.
- Existing systems lack proper real-time communication between patients and nurses.
- There is a need for an automated system that provides scheduled reminders and emergency alerts.
- A reliable RTOS-based solution is required to improve healthcare efficiency and patient care.

OBJECTIVES:

- To design and implement an RTOS-based smart hospital communication system.
- To provide automatic time-triggered medicine reminders for nurses.
- To develop an event-driven alert system for patients to request medicines or assistance.

- To improve real-time communication between patients and nurses.
- To reduce delays and human errors in medicine administration.
- To enhance patient safety and healthcare efficiency.
- To utilize RTOS features such as task scheduling, queues, semaphores, and interrupts.
- To create a reliable and scalable system for smart hospital applications.

CHAPTER 2

LITERATURE SURVEY

Real-Time Operating System (RTOS)-based healthcare monitoring systems have gained significant importance in modern medical applications due to their ability to manage multiple tasks efficiently and provide reliable real-time performance. Researchers have developed various embedded healthcare systems that use RTOS for patient monitoring, medicine scheduling, emergency alert generation, and communication between medical devices. RTOS enables accurate task scheduling, fast response time, and proper resource management, making it suitable for critical healthcare environments where timely operation is essential for patient safety and effective hospital management.

Event-driven communication techniques are widely used in embedded healthcare systems to improve responsiveness during emergencies. In event-driven systems, tasks are executed only when specific events occur, such as patient requests, sensor triggers, or emergency notifications. Studies show that this approach reduces unnecessary processor utilization and allows quick handling of urgent situations. In hospital applications, event-driven communication helps patients notify nurses immediately when assistance or medicines are required, thereby improving nurse-patient interaction and reducing delays in healthcare services.

Time-triggered communication systems are commonly used for periodic monitoring and scheduled operations in hospitals. In this approach, tasks are executed at predefined time intervals using RTOS timers and schedulers. Researchers observed that time-triggered systems improve timing accuracy and ensure regular medicine reminders, vital sign monitoring, and routine healthcare operations. These systems are highly reliable because tasks are performed continuously according to a fixed schedule, reducing the possibility of missed medicine administration and improving patient care quality.

Recent advancements in IoT and wireless communication technologies have further improved smart hospital systems. Researchers have integrated wireless modules, mobile applications, and communication protocols such as MQTT, Wi-Fi, and Bluetooth into healthcare monitoring systems to enable real-time data transfer and remote patient monitoring. Modern systems combine both event-driven and time-triggered approaches to provide flexible and efficient healthcare solutions. From the literature survey, it is observed that integrating both communication methods using RTOS can significantly improve patient safety, reduce human errors, enhance hospital efficiency, and provide reliable real-time communication between patients

and nurses.

CHAPTER 3

PROPOSED METHODOLOGY AND ARCHITECTURE

The methodology is based on integrating Time-Triggered and Event-Driven communication mechanisms using RTOS task scheduling, inter-task communication, and a real-time dashboard monitoring system.

2.1 System Classification

The system is divided into four levels:

- Patient Unit

This unit consists of:

- Push button / alert switch
- Patient communication interface
- Emergency request module

The patient can send medicine requests or emergency alerts to the nurse.

- Nurse Monitoring Unit

This unit is responsible for:

- Receiving medicine reminder notifications
- Receiving patient alert messages
- Monitoring medicine schedules
- Responding to patient requests

- Dashboard Monitoring Unit

This unit provides a real-time monitoring dashboard for hospital staff.

Dashboard Functions:

- Displays patient medicine schedules
- Shows pending medicine alerts
- Displays patient emergency messages
- Tracks nurse acknowledgment status
- Provides real-time communication updates

The dashboard can be implemented using:

- Node-RED dashboard
- Web dashboard
- IoT cloud interface
- Mobile monitoring application
- Control Unit

This unit consists of:

- Microcontroller with RTOS
- Communication module (Wi-Fi/Bluetooth)
- Task scheduler and monitoring system

The control unit manages all real-time operations and communication between patient, dashboard, and nurse units.

2.2 Data Communication Process

- Medicine schedules are predefined in the RTOS scheduler.
- The system continuously monitors medicine timing.
- At scheduled intervals, the system automatically sends reminder notifications to nurses.
- Reminder messages are also displayed on the monitoring dashboard.

- If the patient does not receive medicine on time:
 - The patient presses an alert button or sends a request.
 - The event-driven task is immediately activated.
 - RTOS processes the request and forwards the alert message to:
 - Nurse monitoring unit
 - Dashboard monitoring system
 - The nurse receives the notification and provides medicine to the patient.
 - Dashboard status updates after nurse acknowledgment.
-

2.3 Time-Triggered Communication

The system uses RTOS timers and schedulers for periodic operations.

Functions:

- Automatic medicine reminder generation
- Periodic monitoring of patient schedules
- Dashboard status updates
- Continuous task execution at fixed intervals

Benefits:

- Accurate medicine timing
 - Reduced chances of missed medication
 - Reliable periodic monitoring
 - Centralized monitoring through dashboard
-

2.4 Event-Driven Communication Mechanism

The event-driven system operates whenever an emergency or patient request occurs.

Event Conditions:

- Patient medicine delay
- Emergency assistance request
- Nurse acknowledgment notification
- Dashboard alert generation

Working Process:

- Patient triggers an alert using a push button or communication module.
- RTOS interrupt/event task becomes active immediately.
- Notification message is sent to:
 - Nurse station
 - Dashboard monitoring system
- Dashboard displays real-time emergency alerts.
- The nurse responds to the patient request in real time.

Advantages:

- Faster response time
 - Reduced patient waiting time
 - Efficient emergency handling
 - Real-time dashboard monitoring
-

2.5 RTOS Task Management

The system uses RTOS features for multitasking and synchronization.

Tasks Implemented:

- Medicine Reminder Task
- Patient Alert Task
- Dashboard Update Task
- Communication Task
- Display/Monitoring Task

RTOS Features Used:

- Task scheduling
- Semaphores
- Message queues
- Interrupt handling
- Inter-task communication

These features ensure efficient and reliable real-time system operation.

2.6 Smart Monitoring Mechanism

The system continuously monitors communication between patients and nurses.

If the nurse does not acknowledge the alert:

- The system resends notifications.
- Emergency alerts are displayed on the dashboard.
- Critical warning messages can be generated.
- Communication continuity is maintained.

SMART HOSPITAL MEDICINE REMINDER AND PATIENT ALERT SYSTEM ARCHITECTURE

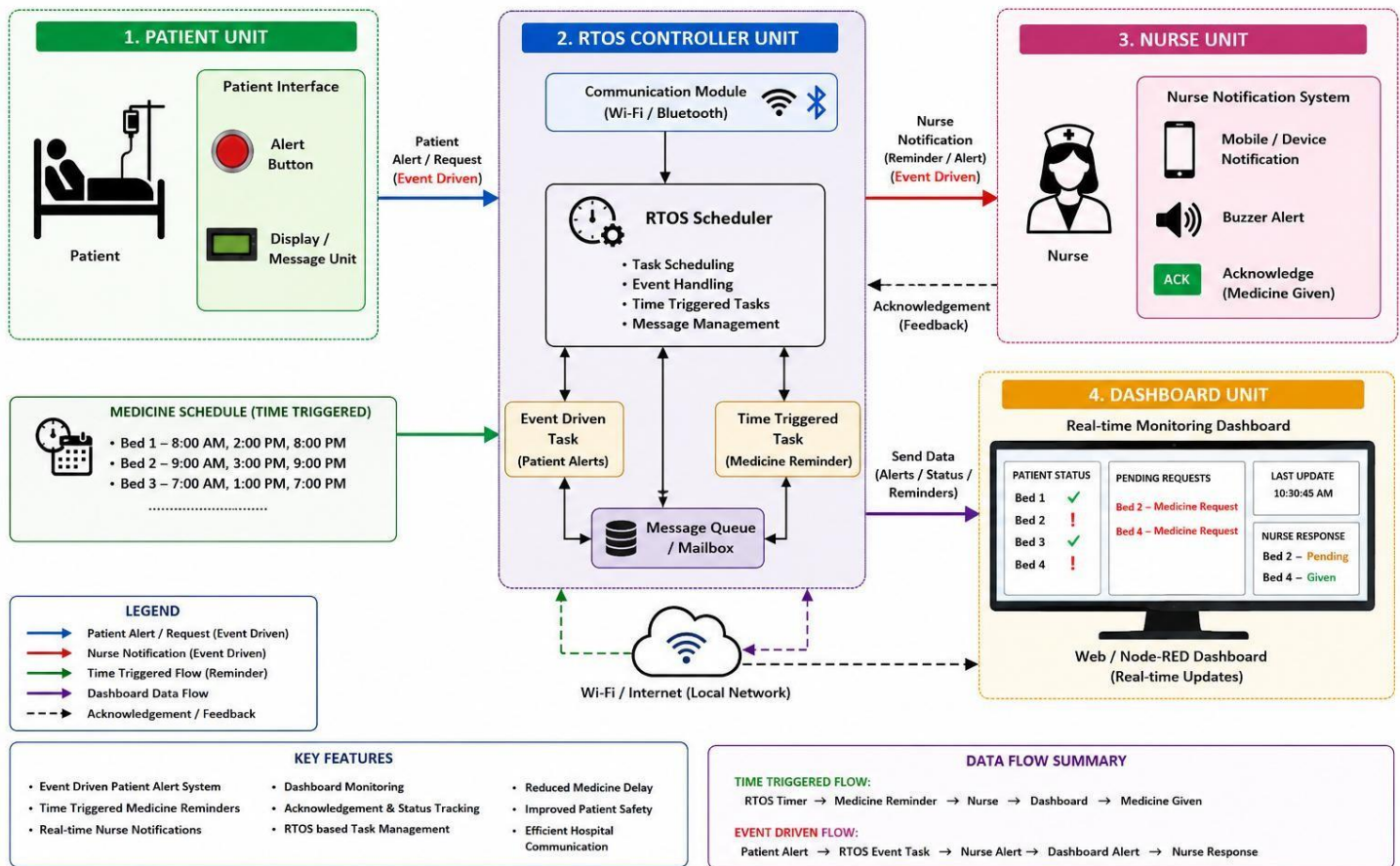


FIG NO 2.1.1 ARCHITECTURE

CHAPTER 4

HARDWARE AND SOFTWARE USED

Hardware Components:

1. ESP32 (Multiple Units)
2. STM32F401RE
3. Jumper Wires & Breadboard
For circuit connections and prototyping.
4. Power Supply (USB / 5V Adapter)
Provides power to ESP32 and STM32 boards.

Software Components:

1. Arduino IDE
Used to program ESP32 nodes (sensor, intermediate, and gateway).
2. STM32CubeIDE
Used to configure and program the STM32 microcontroller.
3. PuTTY
Used to monitor UART output from STM32.
4. Embedded C / Arduino C++
Programming languages used for firmware development.
5. Dash board:
Patient status, Patient requirement, last update, Nurse response

CHAPTER 5

EXPECTED OUTCOME

The proposed system, RTOS-Based Smart Hospital Patient Care System using Event-Driven Communication and Time-Triggered Medicine Reminder, is expected to achieve the following outcomes:

1. Timely Medicine Reminder System

- The STM32 controller running RTOS will continuously monitor predefined medicine schedules.
- Time-triggered tasks will automatically generate medicine reminders at scheduled intervals.
- Notifications will be sent to the nurse unit and dashboard.

Outcome:

- Accurate medicine delivery timing
 - Reduced chances of missed medication
 - Improved patient healthcare management
-

2. Event-Driven Patient Alert System

If the nurse forgets to provide medicine due to workload or emergency situations:

- The patient can press an alert button connected to the ESP32 patient node.
- ESP32 will immediately send the request message through Wi-Fi/MQTT communication.
- RTOS event task will process the request in real time.

Outcome:

- Immediate nurse notification
 - Faster response to patient requests
 - Reduced patient waiting time
-

3. Real-Time Dashboard Monitoring

- All medicine reminders and patient alerts will be displayed on the dashboard.
- Dashboard will continuously update:

 Patient medicine status

 Pending requests

 Nurse acknowledgment

 Emergency alerts

Outcome:

- Centralized hospital monitoring system
- Real-time tracking of patient requests
- Improved hospital management efficiency

Example:

- BED 2 → Medicine Pending
- BED 4 → Emergency Alert
- Nurse Response → Acknowledged

4. Reliable RTOS Task Management

The RTOS system will efficiently manage:

- Time-triggered reminder tasks
- Event-driven patient request tasks
- Dashboard update tasks
- Communication tasks

Using:

- Task scheduling
- Message queues
- Semaphores
- Interrupt handling

Outcome:

- Reliable multitasking operation
- Real-time communication handling
- Reduced processing delay

5. ESP32 Wireless Communication

- ESP32 nodes will provide wireless communication between:
 - Patient unit
 - Nurse unit
 - Dashboard system
- MQTT/Wi-Fi communication will transfer alert messages and status updates.

Outcome:

- Stable wireless communication
- Real-time data transfer
- Smart IoT-based hospital system

6. Nurse Acknowledgment and Feedback System

- After receiving the alert, the nurse can acknowledge the request.
- The acknowledgment status will be updated on the dashboard.

Outcome:

- Better nurse-patient coordination
- Monitoring of completed medicine delivery
- Reduced communication gaps

Example:

- MEDICINE GIVEN → ACKNOWLEDGED
 - ALERT STATUS → COMPLETED
-

7. Emergency Alert Handling

If patient requests are not acknowledged within a predefined time:

- The system will resend notifications.
- Emergency alerts will be activated on the dashboard and buzzer system.

Outcome:

- Improved patient safety
 - Reliable emergency communication
 - Continuous healthcare monitoring
-

8. Smart Hospital Automation

The complete system will integrate:

- ESP32
- STM32
- RTOS
- MQTT/Wi-Fi communication
- Dashboard monitoring

Outcome:

- Smart hospital patient care system
- Automated medicine reminder solution
- Efficient nurse-patient communication
- Scalable and modern healthcare architecture

CHAPTER 6

APPLICATIONS/SCOPE

- **Smart hospitals and healthcare centers** – Used for automated patient monitoring and medicine management.
- **Patient medicine reminder systems** – Helps nurses provide medicines at the correct time.
- **Nurse-patient communication systems** – Enables real-time communication between patients and nurses.
- **Emergency patient alert systems** – Allows patients to send emergency alerts instantly.
- **ICU and critical patient monitoring** – Supports continuous monitoring of critical patients.
- **Elderly care and nursing homes** – Assists elderly patients in receiving timely medical support.
- **Rehabilitation centers** – Improves patient care and medicine scheduling during recovery.
- **IoT-based healthcare monitoring** – Uses wireless communication for smart healthcare applications.
- **Real-time hospital automation systems** – Automates hospital operations using RTOS technology.
- **Wireless patient monitoring applications** – Provides wireless monitoring through ESP32 communication.
- **Remote healthcare assistance systems** – Helps healthcare staff monitor patients remotely.
- **RTOS-based embedded healthcare projects** – Demonstrates real-time embedded system applications in healthcare.

Future Scope

- 1.Integration with cloud platforms (IoT dashboards)
- 2.Implementation of advanced routing algorithms
- 3.AI-based fault prediction
- 4.Mobile app monitoring system
- 5.Energy-efficient node design

CHAPTER 7

PROJECT TIMELINES:

- Requirement Analysis and Research – 2 weeks
- System Design and Hardware Setup – 3 weeks
- Coding and Cloud Integration – 4 weeks
- Testing and Optimization – 2 weeks
- Documentation and Final Presentation – 1 week

CHAPTER 8

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